

DecodeLabs Industrial Training Kit

Batch 2026

PROJECT 2

Exploratory Data Analysis (EDA)

Uncovering the Story — From Descriptive Statistics to Business Diagnosis

[This report was prepared as part of the DecodeLabs Industrial Training Programme, Data Analytics track. It documents the exploratory analysis of the dataset cleaned and verified in Project 1, and establishes the analytical findings carried into Projects 3 and 4.]

Dataset Analysed

E-Commerce Sales Dataset (Project 1 verified output) — 1,200 orders × 14 columns

Order Period: January 2023 — June 2025 (30 months)

Tool: Microsoft Excel

Submitted to DecodeLabs | July 2026

Executive Summary

41.4% of all order value is at risk — cancellations and returns, not customer acquisition, are the biggest revenue lever in this business. That is the headline finding of this exploratory analysis of 1,200 verified e-commerce orders (£1.26M gross order value) spanning January 2023 to June 2025.

The analysis ran in five phases: descriptive statistics, category breakdowns, trend analysis, outlier detection, and correlation analysis, with every finding passed through a business-impact (“So What?”) test. The distribution of order values is right-skewed — the mean order (£1,053.97) sits 28% above the median (£823.62) — so the median is used as the reliable measure of a typical order throughout. Revenue is evenly spread across all seven product categories (each holding 12–15.5% of revenue), and monthly revenue is flat, fluctuating around £42K with no sustained growth across the 30-month window.

The most consequential findings: 497 orders (41.4%) ended Cancelled or Returned, putting £519,674 of gross value at risk — more than the £488,760 realised through Delivered and Shipped orders. Instagram is simultaneously the highest-volume channel (259 orders) and the most reliable (15.8% cancellation rate), while Google (24.1%) and Email (23.6%) leak the most orders. Coupons show no effect on basket size. Outlier analysis (IQR method) found 8 high-value orders above the upper bound — all legitimate maximum-quantity purchases rather than data errors, marking a small premium segment worth studying. A moderate positive correlation ($r = 0.65$) between cart size and quantity purchased suggests cart-building features can lift order size.

Table of Contents

Section 1: Methodology & Framework..... 1

 1.1 The EDA Framework 1

 1.2 Data Source & Scope 1

 1.3 Key Techniques Applied..... 1

Section 2: Phase 1 — Descriptive Statistics..... 2

Section 3: Phase 2 — Category Breakdowns..... 3

 3.1 Revenue by Product..... 3

 3.2 Performance by Referral Source 3

 3.3 Spending by Coupon Code 4

 3.4 Orders by Payment Method..... 5

 3.5 Order Status Mix — Realised vs At-Risk Revenue 5

Section 4: Phase 3 — Trend Over Time 7

Section 5: Phase 4 — Outlier Analysis (IQR Method)..... 8

Section 7: Key Observations — The “So What?” Test 9

Section 8: Recommendations10

Section 9: Conclusion.....10

Summary Results Table11

Section 1: Methodology & Framework

1.1 The EDA Framework

The analysis treats the dataset as a repository of evidence and follows the Input–Process–Output structure: the evidence (the verified Project 1 dataset), the investigation (mathematical transformations and statistical logic), and the verdict (actionable insights and decision support). The work is organised in five phases:

- **Phase 1:** Descriptive statistics — count, mean, median, min, max, standard deviation, and quartiles for every key numeric field.
- **Phase 2:** Category breakdowns — orders, revenue, average order value, and revenue share by product, referral source, coupon code, payment method, and order status.
- **Phase 3:** Trend over time — monthly orders, revenue, and average order value across the full 30-month window.
- **Phase 4:** Outlier analysis — the IQR method applied to order value, with each flagged order individually inspected to separate signal from noise.
- **Phase 5:** Key observations — every statistical finding translated into a business implication before being recorded.

1.2 Data Source & Scope

The input is the cleaned, verified dataset produced by Project 1: 1,200 e-commerce order records × 14 columns, with zero duplicate IDs, zero incorrectly formatted dates, zero blank cells, and zero calculation mismatches. Because the data passed the Project 1 verification gate, every pattern found in this analysis can be attributed to genuine business behaviour rather than data-quality artefacts. The raw backup, cleaned dataset, and Project 1 cleaning log travel inside the same workbook for full traceability.

1.3 Key Techniques Applied

- **Descriptive statistics:** COUNT, AVERAGE, MEDIAN, MIN, MAX, STDEV, and QUARTILE across TotalPrice, UnitPrice, Quantity, and ItemsInCart.
- **Mean-vs-median diagnosis:** comparing the mean against the median to detect right-skew and selecting the robust measure for each use case.
- **Pivot-style breakdowns:** aggregation of orders, revenue, average order value, revenue share, and cancellation rates by category.
- **Trend analysis:** monthly grouping of orders and revenue across 30 months to test for growth, decline, and seasonality.
- **IQR outlier detection:** $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ bounds on order value, chosen over the Z-score method because IQR is robust for skewed business data.
- **Correlation analysis:** Pearson correlation (CORREL) between ItemsInCart and Quantity, interpreted as a clue rather than a verdict (correlation $\neq$ causation).

Section 2: Phase 1 — Descriptive Statistics

The five-number summary plus mean and standard deviation was computed for the four key numeric fields:

Statistic	TotalPrice (£)	UnitPrice (£)	Quantity	ItemsInCart
Count	1,200	1,200	1,200	1,200
Mean	1,053.97	356.41	2.95	5.49
Median	823.62	364.21	3	5
Min	11.39	11.39	1	1
Max	3,456.40	699.93	5	10
Std Dev	819.86	197.18	1.41	2.28
Q1 (25th pct)	410.52	186.06	2	4
Q3 (75th pct)	1,578.48	521.57	4	7

The defining feature of the distribution is the gap between the mean order value (£1,053.97) and the median (£823.62) — a difference of £230.35, or 28%. The mean sits well above the median because a minority of large orders pulls the average upward: the distribution of order values is right-skewed. Unit prices, by contrast, are close to symmetrical (mean £356.41 vs median £364.21), which localises the skew to the interaction of quantity and price at the order level rather than to the price list itself.

The practical consequence, applied throughout this report: the median is the reliable measure of a typical order. Any forecast or per-customer expectation built on the mean would overstate typical behaviour by roughly 28%.

Section 3: Phase 2 — Category Breakdowns

3.1 Revenue by Product

Product	Orders	Revenue (£)	Avg Order Value (£)	% of Revenue
Chair	178	195,620.11	1,098.99	15.5%
Printer	181	195,612.61	1,080.73	15.5%
Laptop	173	192,126.56	1,110.56	15.2%
Tablet	179	186,568.95	1,042.28	14.8%
Monitor	163	175,651.41	1,077.62	13.9%
Desk	170	167,459.93	985.06	13.2%
Phone	156	151,722.39	972.58	12.0%

Chair leads revenue (£195,620) with Printer nearly tied (£195,613); Phone is lowest (£151,722). Every product sits between 12% and 15.5% of revenue — revenue is evenly spread, not concentrated. There is no single-product dependency risk, but equally no star product driving growth.

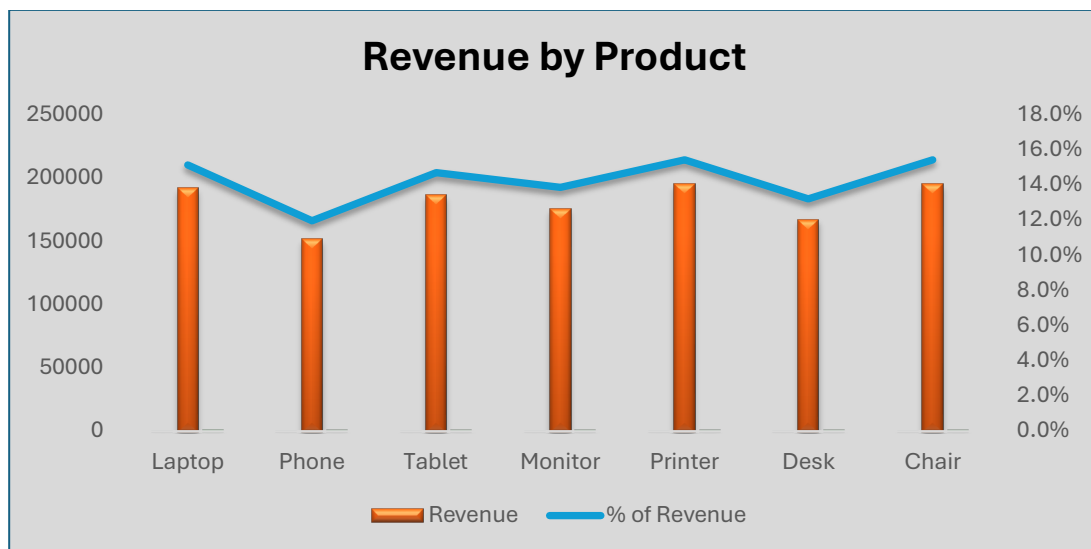


Figure 1: Revenue by product — orders (bars) and revenue (line).

3.2 Performance by Referral Source

Channel	Orders	Revenue (£)	Avg Order (£)	% of Rev	Cancellation Rate
Instagram	259	275,285.45	1,062.88	21.8%	15.8%
Email	250	261,808.55	1,047.23	20.7%	23.6%
Google	241	250,441.48	1,039.18	19.8%	24.1%
Facebook	228	250,410.90	1,098.29	19.8%	18.4%
Referral	222	226,815.58	1,021.69	17.9%	22.5%

Instagram is the standout channel on both dimensions that matter: it brings the most orders (259) and loses the fewest of them (15.8% cancellation rate). Google carries the highest cancellation rate (24.1%) with Email close behind (23.6%) — nearly one in four orders from those channels falls through.

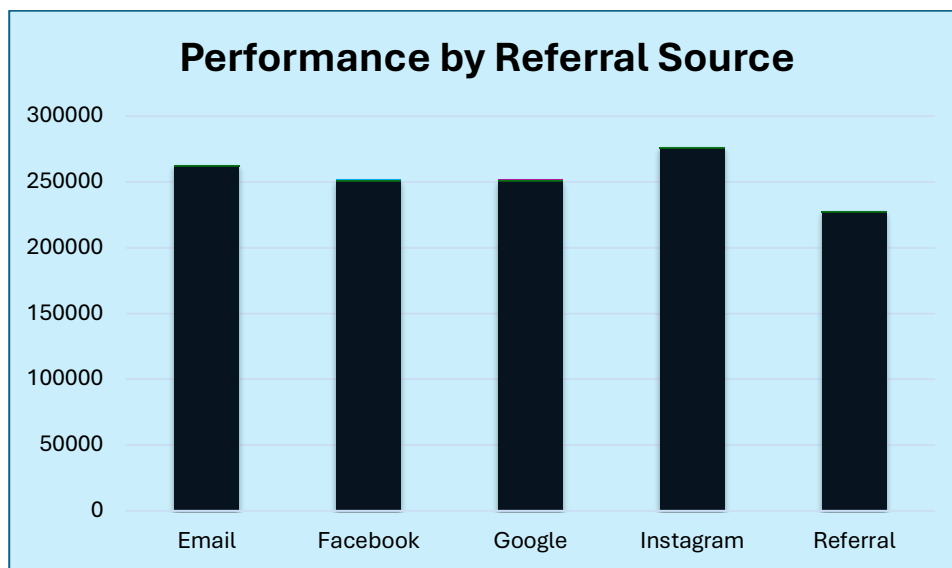


Figure 2: Performance by referral source

3.3 Spending by Coupon Code

Coupon	Orders	Revenue (£)	Avg Order Value (£)
FREESHIP	313	335,036.99	1,070.41
NO_COUPON	309	322,401.41	1,043.37
WINTER15	292	302,483.54	1,035.90
SAVE10	286	304,840.02	1,065.87

Average order value is nearly identical across all four groups — coupon users spend no more per order than NO_COUPON customers (the segment created by the Project 1 imputation). Coupons are not increasing basket size; unless they demonstrably drive acquisition or retention, they function as margin giveaways.



Figure 3: Spending by coupon code

3.4 Orders by Payment Method

Payment methods are evenly used — orders range only from 230 (Gift Card) to 258 (Online). Credit Card carries the highest average order value (£1,127.55) and Debit Card the lowest (£1,001.56), a narrow spread indicating no payment-method effect worth acting on.

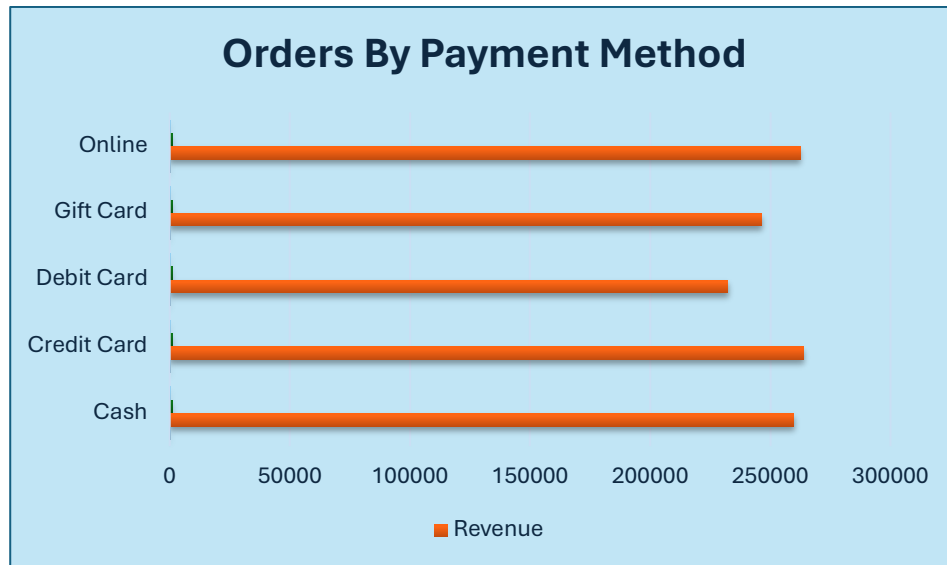


Figure 4: Orders by payment method

3.5 Order Status Mix — Realised vs At-Risk Revenue

Status	Orders	% of Orders	Gross Value (£)
Cancelled	250	20.8%	276,396.21
Returned	247	20.6%	243,277.70
Pending	237	19.8%	256,328.15
Shipped	235	19.6%	246,159.58
Delivered	231	19.3%	242,600.32

This is the single most consequential finding in the analysis. 497 orders — 41.4% of everything sold — ended Cancelled or Returned, placing £519,674 of gross order value at risk. That exceeds the £488,760 actually realised through Delivered and Shipped orders (38.8%): the business currently loses more order value than it captures.

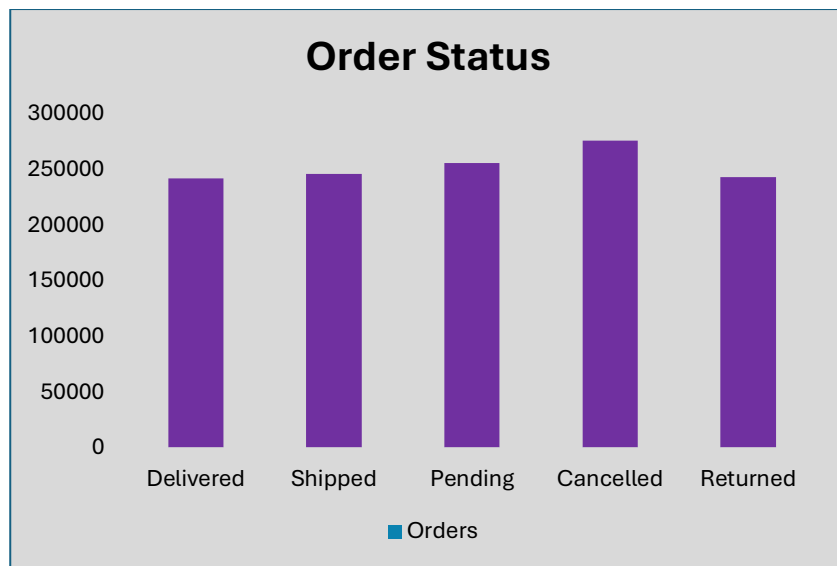


Figure 5: Order status mix

Section 4: Phase 3 — Trend Over Time

Orders and revenue were grouped by month across the full 30-month window. The year-level summary:

Year	Months	Orders	Revenue (£)	Avg Monthly Revenue (£)
2023	12	510	552,643.24	46,053.60
2024	12	459	480,235.87	40,019.66
2025 (H1)	6	231	231,882.85	38,647.14

Monthly revenue fluctuates around £42K with no sustained growth across the period — the strongest month was June 2024 (£68,069) and the weakest April 2023 (£27,752), but the swings are noise around a flat baseline rather than a trend. If anything, the yearly averages drift slightly downward (£46.1K → £40.0K → £38.6K per month). No consistent seasonal spike repeats across the two full years.

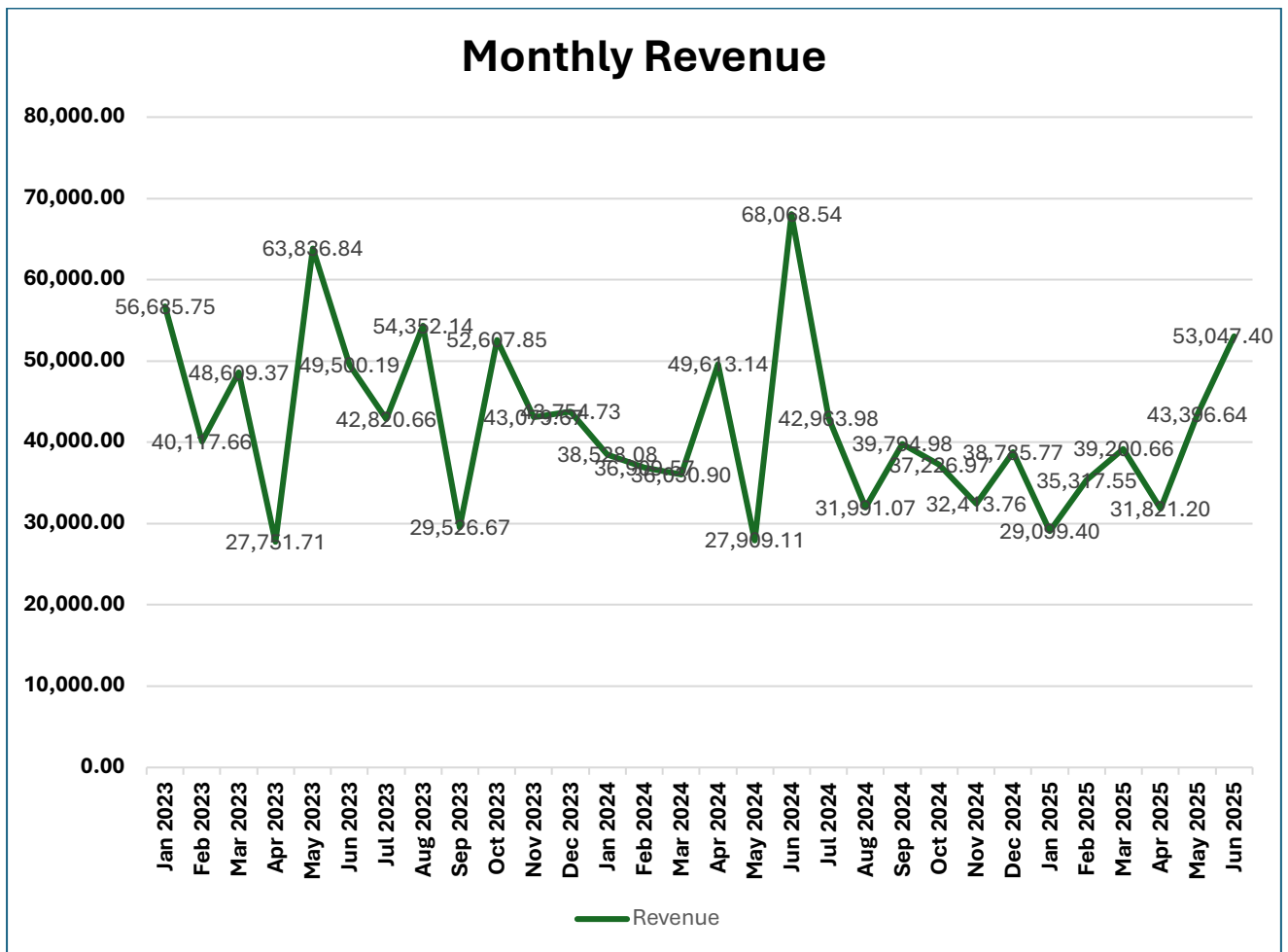


Figure 6: Monthly Revenue Trend

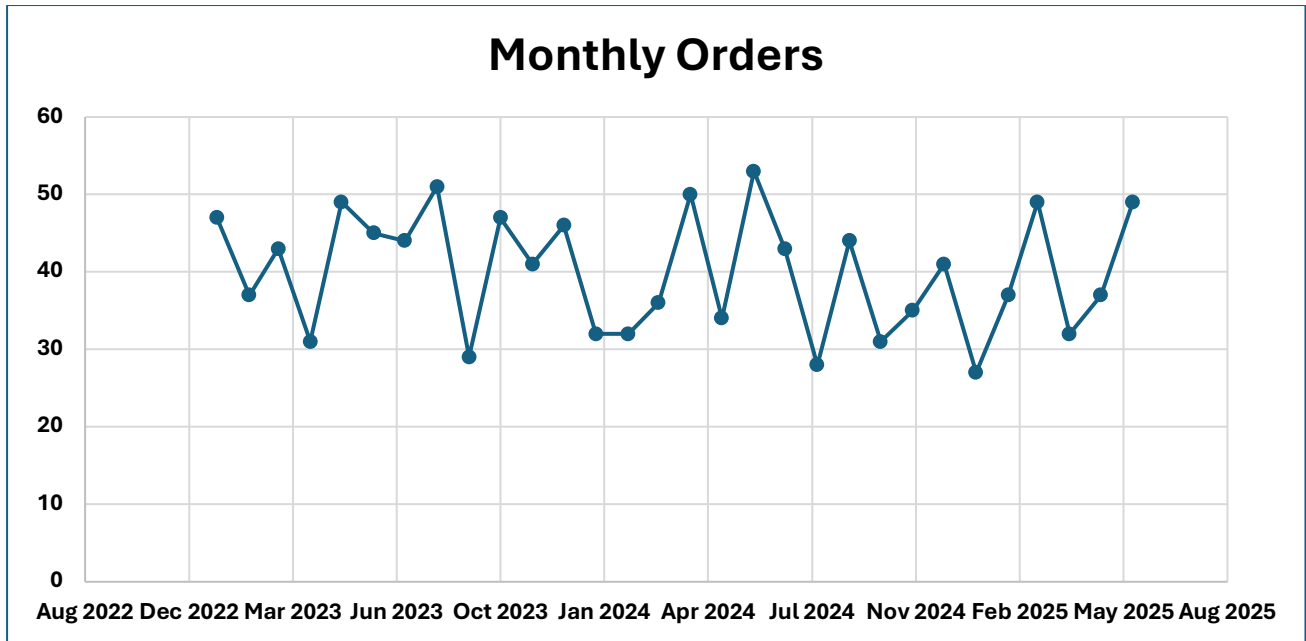


Figure 7: Monthly order counts over the period

The business implication is direct: the business is stagnant. Growth will not come from riding an existing trend — it must be engineered, and the two levers this analysis identifies are fixing the cancellation problem (Section 3.5) and reallocating channel investment (Section 3.2).

Section 5: Phase 4 — Outlier Analysis (IQR Method)

The IQR method was applied to TotalPrice. It was chosen over the Z-score method because IQR is robust against extreme values, making it the appropriate tool for skewed business data (the right-skew was established in Section 2).

Parameter	Value
Q1 (25th percentile)	£410.52
Q3 (75th percentile)	£1,578.48
IQR (Q3 – Q1)	£1,167.96
Lower bound (Q1 – 1.5 × IQR)	–£1,341.41
Upper bound (Q3 + 1.5 × IQR)	£3,330.41
Orders below lower bound	0
Orders above upper bound	8

Eight orders exceed the upper bound, led by ORD200789 at £3,456.40; none fall below it (the lower bound is negative, which no valid order can breach). Each of the eight was inspected individually — the signal-vs-noise test: all are legitimate maximum-quantity (5-unit) purchases of high-priced products, not data-entry errors. Under the framework's classification these outliers are signal, not noise: nothing is removed, and the eight orders mark a small high-value customer segment worth studying rather than a cleaning problem.

Section 6: Phase 5 — Correlation Analysis

Pearson correlation between ItemsInCart and Quantity purchased returned $r = 0.65$ — a moderate positive relationship. Customers who build larger carts go on to buy larger quantities.

Applying the golden rule that correlation is a clue, not a verdict: this does not prove that cart size causes larger purchases — a confounder such as customer intent or basket type could drive both. But the strength of the relationship justifies a low-cost experiment: cart-building features (bundles, “frequently bought together” prompts) are the natural intervention to test, since if even part of the relationship is causal, they lift order size directly.

Section 7: Key Observations — The “So What?” Test

Every finding was translated from a statistical statement into a business diagnosis before being recorded — the framework's “So What?” test. The eight observations:

#	Theme	What the Data Says	Business Diagnosis
1	Revenue scale & typical order	£1.26M gross value across 1,200 orders; mean order (£1,053.97) is 28% above the median (£823.62) — right-skewed distribution.	The median is the reliable figure for forecasting typical customer behaviour; the mean overstates expected revenue per order.
2	Best & worst products	Chair top (£195,620), Printer nearly tied; Phone lowest (£151,722). All 7 products hold 12–15.5% of revenue.	Revenue is evenly spread — no single-product dependency, but no star product either.
3	Trend over time	Monthly revenue fluctuates around ~£42K with no sustained growth or decline across Jan 2023 – Jun 2025.	The business is stagnant; growth must come from fixing cancellations or channel investment.
4	Order fulfilment risk	41.4% of orders are Cancelled (250) or Returned (247) — £519,674 at risk vs £488,760 realised.	The business loses more order value than it captures; reducing cancellations is a bigger revenue lever than acquiring new orders.
5	Referral source quality	Instagram: most orders (259) and lowest cancellation rate (15.8%). Google: highest cancellation rate (24.1%); Email close behind (23.6%).	Shift spend toward Instagram-like channels; investigate why Google and Email orders fall through.
6	Coupon behaviour	Average order value nearly identical across SAVE10 (£1,065.87), WINTER15 (£1,035.90), FREESHIP (£1,070.41), NO_COUPON (£1,043.37).	Coupons aren't increasing basket size — they're margin giveaways unless they drive acquisition or retention.
7	Outliers	8 orders exceed the IQR upper bound (£3,330.41), max £3,456.40; 0 below. All are legitimate max-quantity × high-price purchases.	No anomalies to remove; the 8 orders mark a small high-value segment worth studying.
8	Correlation	$\text{CORREL}(\text{ItemsInCart}, \text{Quantity}) = 0.65$ — a moderate positive relationship.	Larger carts convert into larger purchases; cart-building features (bundles, “frequently bought together”) likely lift order size.

Section 8: Recommendations

The recommendations below follow directly from the business diagnoses in Section 7:

- **1. Make cancellation reduction the top priority:** 41.4% of order value is lost to cancellations and returns — a larger pool than realised revenue itself. Diagnosing and reducing this leakage is worth more than any acquisition campaign of realistic size.
- **2. Reallocate channel investment:** Instagram delivers the highest volume at the lowest cancellation rate; Google and Email lose roughly one order in four. Shift marketing spend toward Instagram-like channels and investigate the failure points in the Google and Email journeys.
- **3. Review the coupon programme:** Coupons show no effect on basket size across any code. Unless attribution data shows they drive acquisition or retention, they are pure margin giveaways and should be tested against a no-coupon control.
- **4. Study the high-value segment:** The 8 orders above the IQR upper bound are legitimate premium purchases. Profile these customers and orders to understand what drives maximum-quantity, high-price buying.
- **5. Test cart-building features:** The $r = 0.65$ cart-quantity correlation justifies piloting bundles and “frequently bought together” prompts — a low-cost experiment with a plausible causal path to larger orders.
- **6. Forecast on the median:** Use the median (£823.62), not the mean, for any forecast of typical order value; reserve the mean for total-revenue arithmetic only.

Section 9: Conclusion

This analysis transformed a static table of 1,200 verified orders into a clear business diagnosis: a stagnant, evenly-spread business whose biggest problem — and biggest opportunity — is that it loses more order value to cancellations and returns than it realises in completed sales. The supporting evidence is consistent across every phase: flat monthly revenue, no product concentration, one clearly superior marketing channel, an ineffective coupon programme, a small legitimate premium segment, and a cart-size relationship worth exploiting.

Because the input dataset passed the Project 1 verification gate, each of these patterns reflects genuine business behaviour rather than data artefacts. The findings established here become the reference figures for Project 3, where the same questions are re-answered in SQL and cross-validated against these EDA results, and the analytical narrative for the Project 4 Power BI dashboard, where these insights are amplified through visualisation.

Summary Results Table

The table below consolidates every key metric from this analysis into a single reference.

Metric	Result
Orders analysed	1,200 (verified Project 1 output)
Total gross order value	£1,264,761.96
Mean / median order value	£1,053.97 / £823.62 (28% gap → right-skewed)
Order value range	£11.39 – £3,456.40
Top / bottom product by revenue	Chair £195,620 / Phone £151,722 (all 12–15.5% share)
Monthly revenue trend (30 months)	Flat, ~£42K/month; no seasonality
Orders Cancelled or Returned	497 (41.4%) — £519,674 at risk
Realised revenue (Delivered + Shipped)	£488,760 (38.8% of orders)
Best channel	Instagram — 259 orders, 15.8% cancellation rate
Weakest channels	Google (24.1%) and Email (23.6%) cancellation rates
Coupon effect on basket size	None — AOV nearly identical across all codes
Outliers (IQR method)	8 above upper bound (£3,330.41); all legitimate; 0 below
Correlation: ItemsInCart vs Quantity	$r = 0.65$ (moderate positive)
Tool	Microsoft Excel

Analyst: Rubel Hasan · DecodeLabs Industrial Training Programme · Data Analytics Track · Batch 2026. This report was prepared as part of a training programme; the dataset was provided by DecodeLabs for educational purposes. Monetary figures are gross order values from the training dataset.